

Predicting TF Binding Potential of DNA Sequences

Course Project Report

BI3464/DS3414 - Computational Functional Genomics

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	CTCF Markov	CTCF exp	CTCF actual	REST Markov	REST exp	REST actual	EP300 Markov	EP300 exp	EP300 actual
auROC	0.739	0.898	0.896	0.783	0.908	0.905	0.778	0.92	0.922
auPRC	0.039	0.39	0.400	0.053	0.217	0.228	0.023	0.128	0.126
Best Markov Order	5	-	-	5	-	-	4	-	-

Table 1: **Final performance of models**

1 Models Overview

1.1 CTCF Model

PhastCons + FIMO + ATAC + JASPAR + SVM

1.2 REST Model

k-mer + MARKOV + ATAC + Logistic Regression

1.3 EP300 Model

JASPAR proxy PWMs + Markov + GC + ATAC + Logistic Regression

2 Data Sources

2.1 Primary Data

- Chromosome-level TSV files (chr1–chr22, excluding chr3, chr10, chr17): each file contains columns chr, start, end, ATAC, CTCF, REST, EP300.
- Reference genome: hg38.fa - used to extract 200 bp DNA sequences for each labelled bin via pyfaidx.

2.2 External Data

1. JASPAR database (<https://jaspar.elixir.no/>) - Position Frequency Matrices (PFMs) for:
 - (a) CTCF: MA0139.2
 - (b) REST: canonical 20-mer motif (MA0138.2)
 - (c) EP300 proxy TFs: NFIX, AP1, ETS, GATA, FOX, RUNX, TEAD
2. PhastCons 30-way conservation scores for hg38 — bigWig file downloaded from UCSC Genome Browser (hg38.phastCons30way.bw). Used only in the CTCF pipeline.
3. MEME motif files — .meme format conversions of JASPAR PFMs, used as input to the FIMO tool (MEME Suite). Used only in the CTCF pipeline.

2.3 Data Processing

- All .tsv files were loaded with pandas; columns were normalised to canonical names (chr, start, end, ATAC, TF).
- DNA sequences were extracted from hg38.fa using pyfaidx; regions containing 'N' bases were excluded from training.
- The ATAC column was treated as a binary accessibility signal: B $\rightarrow$ 1, U $\rightarrow$ 0 (or 0.02 in the SVM pipeline)
- For training, bound (B) and unbound (U) regions were sub-sampled to manage class imbalance: up to 10,000 bound and up to 100,000 unbound sequences (10:1 ratio cap), with class_weight='balanced' in all classifiers.

3 Methods Tried

3.1 Baseline: Markov Model (Midterm Goal)

This model scores sequences by computing the log-odds ratio of the probability under a 'bound' Markov model versus an 'unbound' Markov model, where each model is trained on the respective set of sequences from the training folds.

- Results: AUROC $\sim$ 0.74–0.78 across TFs; AUPR very low ($\sim$ 0.02–0.05) due to severe class imbalance ($\sim$ 95% unbound).

3.2 Methods That Did Not Work

Several approaches were explored and discarded due to poor performance or computational constraints:

3.2.1 Fixed-weight feature combination (JASPAR + ATAC + GC + Markov)

Manually weighted combination of PWM score, GC content, ATAC status and Markov log-odds. REST (with elevated GC content weight) and CTCF were tested. Manual weights were inferior to learned weights; the model was unable to adapt to the different sequence architectures across TFs.

3.2.2 Single JASPAR PWM for EP300

EP300 is a histone acetyltransferase co-activator that does not bind DNA directly. Using a single PWM for EP300 with logistic regression failed because there is no canonical EP300-DNA motif in JASPAR. The model effectively learned nothing beyond baseline.

3.2.3 MEME de novo motif discovery + FIMO

MEME was run on bound sequences to discover de novo motifs, which were then used with FIMO for genome-wide scanning. AUPR remained at baseline ($\sim$ 0.05) because MEME found no consistent enriched motifs across the highly variable bound regions, and FIMO hit fractions were low at a strict p-value threshold 0.0001 for 200bp bins, suggesting no real binding signal was captured.

3.2.4 z-score combined SVM and k-mer feature vector (full k=4, 256 features)

A raw k-mer frequency matrix (all 4-mers, 256 columns) was built and concatenated with ATAC and Markov scores before logistic regression. This approach blew up RAM on the available laptops when scaled to all 19 training chromosomes, making it impractical. Also, it z-scored all features into a single scalar before passing to the SVM. This made features redundant — individual feature informativeness was lost in the scalar collapse. Switched to separate z-scored features instead.

4 Final Methods (Best Performing)

4.1 CTCF: JASPAR PWM + PhastCons + FIMO + ATAC + Linear SVM

4.1.1 Features (4-dimensional, separately z-scored)

- PWM score: best log2-likelihood score for the CTCF JASPAR motif MA0139.2 across all positions (both strands) of the 200 bp bin.
- PhastCons conservation score: mean vertebrate conservation score over the region of the best PWM hit, from the hg38.phastCons30way.bw bigWig file using pyBigWig.
- ATAC accessibility: binary signal $B \rightarrow 1.0$, $U \rightarrow 0.02$ (small non-zero to avoid numerical issues).
- FIMO score: $-\log_{10}$ (p-value) of the best FIMO hit from the CTCF MEME motif file, at p-value threshold 0.01. Regions with no hit receive a score of 0.

4.1.2 Classifier

- Linear SVC with balanced class weights, $C=1.0$, $\text{max_iter} = 10,000$.
- Wrapped in CalibratedClassifierCV (sigmoid, $\text{cv} = 3$) to produce calibrated probability outputs.
- All four features z-scored per fold fitted on training data.

4.2 REST: Strand-Invariant k-mers + Markov (m=6) + ATAC + Logistic Regression

4.2.1 Features

- Strand-invariant k-mer frequencies ($k=4$, 136 canonical k-mers): each 200 bp sequence is decomposed into overlapping 4-mers; each k-mer is mapped to the lexicographically smaller of itself and its reverse complement, producing a strand-collapsed frequency vector.
- Markov log-odds score (order $m=6$): log-odds of the sequence probability under a 6th-order Markov model trained on bound sequences versus one trained on unbound sequences, accumulated across all positions.
- ATAC accessibility: binary 0/1.
- Best Markov order selected during CV by maximising mean AUROC across folds: $m=6$ consistently optimal for REST.

4.2.2 Classifier

- Logistic Regression with $\text{max_iter}=2000$.
- Features scaled with z-score.

4.3 EP300: JASPAR Proxy PWMs + Markov + GC Content + ATAC + Logistic Regression

4.3.1 Rationale for proxy PWMs

- The JASPAR proxy PWMs were chosen to represent the major families of TFs known to recruit EP300: AP1 (enhancer pioneer), ETS (widely active), GATA (blood-lineage in K562), FOX (Forkhead pioneer factors), RUNX (blood TF, highly relevant to K562), TEAD (Hippo pathway), and NFIX (nuclear factor). Their combined motif scores act as a proxy for EP300 recruitability.

4.3.2 Features

- Proxy motif score: sum of the best PWM scores across all 7 proxy motifs for each sequence (both strands). Aggregation by sum was found to be more robust than max, because EP300 recruitment is combinatorial.
- Markov log-odds score (order $m=3$ or $m=4$, selected by CV)
- GC content: fraction of G+C bases in the 200 bp sequence. Open chromatin EP300 binding sites tend to be GC-rich enhancers.
- ATAC accessibility: binary 0/1.

4.3.3 Classifier

- Logistic Regression with balanced class weights, `max_iter=2000`.

5 Cross-Validation Design

For the final pipeline, chromosome-level cross-validation was adopted. All 19 training chromosomes are grouped into $k=5$ folds; each fold uses one group of chromosomes as validation and the remaining chromosomes as training. This tests genuine chromosomal generalisation; the model is evaluated on chromosomes it has never seen, which better predicts performance on the held-out chromosomes 3, 10, and 17. (Note: we changed the CV for the EP300 model when running it for this report, since we had to run it for all 3 TFs and leaving one chromosome out wasn't feasible with regards to time available)

6 Results Summary

- AUPR values appear modest in absolute terms because $\sim 95\%$ of all 200 bp bins are unbound (label U), making the baseline precision (random classifier) approximately 5%. The CTCF SVM pipeline produced the best absolute AUPR (0.40), while EP300 remained the hardest TF owing to its indirect binding mechanism and cell-type-specific recruitment.
- Across all three TFs, ATAC accessibility was consistently the most informative feature, confirming the biological principle that chromatin must be open for a TF to bind.
- CTCF: Has a strong, highly conserved 15-mer binding motif. Conservation (PhastCons) adds discriminative power because CTCF binding sites are functionally constrained across vertebrates.
- REST: Prefers GC-rich sequences and has a 20-mer motif with long-range context dependency. The combination of k-mer composition and the Markov model captures both local composition bias and positional sequence dependency.
- EP300: ATAC is critical because EP300 marks active enhancers which are in open chromatin.

Insights from each model for each tf:

	JASPAR PWM + PhastCons+ FIMO + ATAC + Linear SVM	Strand-Invariant k-mers + Markov (m=6) + ATAC + Logistic Regression	JASPAR Proxy PWMs + Markov + GC Content + ATAC + Logistic Regression
CTCF	Mean auPRC - 0.3986 Strong motif, conservation adds to the score	Mean auPRC - 0.264 PWM will have more importance compared to context dependency	Mean auPRC - 0.255 Conservation scores are absent, GC content do not add value here
REST	Mean auPRC - 0.1784 Positional dependency required as motif is not as conserved compared to CTCF	Mean auPRC - 0.217 local composition bias and positional sequence dependency	Mean auPRC - 0.191 GC content adds value compared to other TFs, but PWMs add less value compared to the Kmer model
EP300	Mean auPRC - 0.1123 Proxy PWM taken for only NFIX, which binds predominantly in chr19, so doesn't add value here	Mean auPRC - 0.193 No canonical TF, so the model built from local sequences in the provided data helps. Not enough time to run the code on the submission day	Mean auPRC - 0.14 Multiple proxy PWMs and GC content might add to the PRC value compared to the CTCF model

Table 2: **auPRC and insights of all three models**

Logistic regression score for each feature from the jasper +markov+ gc+atac model:

	CTCF	REST	EP300
Markov	2.503	2.554	2.547
Motif	0.722	0.307	0.206
GC Content	-2.195	-2.010	-2.075
ATAC	1.019	1.186	1.367

Table 3: **Logistic regression coefficients**

Markov and ATAC combined are the most important features in this model (Tab. 3). Motif is very important for CTCF compared to motifs for EP300 and REST. GC content might be capturing the background genome bias, which is why it shows reduced odds of binding

7 Future Improvements

7.1 General

- Replacing the binary ATAC signal with a continuous ATAC read-depth bigWig signal.
- Using gradient boosting (XGBoost/LightGBM) or a shallow neural network instead of logistic regression / linear SVM to capture non-linear feature interactions.
- Distance to nearest TSS as a feature — enhancers and promoters have different binding propensity profiles.
- Model-specific improvements have been mentioned in the presentation

7.2 DNA Shape as an Additional Feature

PWMs model binding purely from base identity at each position, implicitly assuming independence between positions. But because shape is determined by dinucleotide and trinucleotide context, shape parameters naturally capture inter-position dependencies that PWMs miss. Rohs et al. (2009) showed that many TFs recognise the minor groove width profile of their binding site as a shape signature, independently of direct base contacts. Subsequent work demonstrated that adding DNA shape features to PWM scores consistently improves classification of TF-bound versus unbound sequences.

For the three TFs in this project, shape features are biologically motivated:

- CTCF: makes extensive minor groove contacts through its zinc fingers. The narrow minor groove at AT-rich positions within the CTCF motif is a strong shape readout signal.
- REST: has a long 20-mer motif that wraps around the DNA and contacts both grooves. Roll and twist patterns along this extended motif encode the helical phasing necessary for multi-zinc-finger binding.
- EP300: does not bind DNA directly, but the TFs that recruit it (GATA, AP1, RUNX) all have documented shape preferences. Including shape features at the proxy motif sites could improve the proxy score.

7.2.1 Data Source

Pre-computed DNA shape parameters for any sequence can be obtained from the DNASHapeR package (Bioconductor) or from the online DNASHape server (<http://rohslab.cmb.usc.edu/DNASHape/>). These tools use a random forest model trained on all-atom molecular dynamics simulations to predict five shape parameters — Roll, Twist, Propeller Twist, Minor Groove Width, and Helix Rise — at each dinucleotide/trinucleotide step in a query sequence, with no external data files required beyond the sequence itself.

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